

Chapter 1: Integers

10 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-INT-C-01 compares, orders and calculates with integers to solve problems.

NAME _____

CLASS _____

CHAPTER

1

1.01 Integers and opposites

1.02 The number line

1.03 Comparing and ordering integers

1.04 Adding integers

1.05 Subtracting integers

1.06 Multiplying integers

1.07 Dividing integers

1.08 Order of operations with integers

1.09 Integers on a calculator

1.10 Integer problems

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

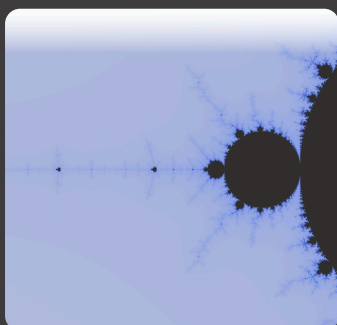
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	7		☐
1.03	Comparing and ordering integers	11		☐
1.04	Adding integers	15		☐
1.05	Subtracting integers	19		☐
1.06	Multiplying integers	23		☐
1.07	Dividing integers	27		☐
1.08	Order of operations with integers	31		☐
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1.10	Integer problems	39		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
1.01

Integers and opposites

Date _____

Learning intentions

We are learning to:

- understand what an integer is
- write the opposite of an integer

Success criteria

I can:

- ☐ describe a real situation with a positive or negative number
- ☐ write the **opposite** of any integer
- ☐ explain why 0 is its own opposite

Key terms Write the definition as your teacher explains it.

Integer

Negative number

Opposite

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write an integer for each situation, or its opposite.

- a** Write an integer for: a temperature 6°C below zero; a gain of 4 metres.

Your teacher works this out. Copy every step.

- b** Write the opposite of: -9 , 15 , 0

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER **INTEGERS** are whole numbers: $\dots, -2, -1, 0, 1, 2, \dots$

ROUND 1 Questions 1–16: write the opposite of each integer.

1 55

2 27

3 45

4 -6

5 -21

6 -19

7 3

8 48

9 21

10 42

11 -39

12 -10

13 -51

14 31

15 16

16 -7

YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 17–28: write an integer for each situation.

- | | | |
|---------------------|----------------------|-------------------------|
| 17 87 steps forward | 18 a rise of 30 cm | 19 50 minutes early |
| 20 69 minutes late | 21 a rise of 61 cm | 22 a loss of \$12 |
| 23 a gain of 43 kg | 24 a deposit of \$65 | 25 a deposit of \$28 |
| 26 67 floors up | 27 73 minutes early | 28 36 m above sea level |

 Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use integers to describe the movement.

- a** A lift goes 3 floors down, then 3 floors up. Write both moves as integers. Where does it end?
- b** Which is further from 0: -8 or 6 ? Explain using opposites.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Negative numbers are below zero.

ROUND 1

Questions 1–16: simplify. The opposite of the opposite is the number itself.

- | | | | |
|----------------------------|-----------------------------|----------------------------|-----------------------------|
| 1 $-(-52)$
_____ | 2 $-(-56)$
_____ | 3 $-(-45)$
_____ | 4 $-(-39)$
_____ |
| 5 $-(-2)$
_____ | 6 $-(-55)$
_____ | 7 $-(-20)$
_____ | 8 $-(-11)$
_____ |
| 9 $-(-10)$
_____ | 10 $-(-19)$
_____ | 11 $-(-3)$
_____ | 12 $-(-50)$
_____ |
| 13 $-(-23)$ | 14 $-(-28)$ | 15 $-(-60)$ | 16 $-(-46)$ |

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: write an integer for the situation, then write its opposite.**17** 30 steps back**18** 39 minutes early**19** 83 m above sea level**20** a deposit of \$21**21** 84 m above sea level**22** 65°C below zero**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Use opposites to find the number.**a** The opposite of a number is 25. What is the number?

Your teacher works this out. Copy every step.

b List the integers whose distance from 0 is exactly 7.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER The **OPPOSITE** is the same distance from 0 on the other side.**1** Is -0 different from 0? Explain.**2** The opposite of a number is 18. What is the number?**3** List all integers whose distance from 0 is less than 3.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Write the opposite of -15 .**Level 2**Write an integer for “a drop of 9°C ”, then write its opposite.**Level 3**The opposite of a number is -31 . What is the number? Explain.

 Extension · Stretch yourself Worked example

The opposite of n is written $-n$. If $n = -4$, what is $-n$?

1. $-n$ means “the opposite of n ”
2. The opposite of -4 is 4

Answer: $-n = 4$

YOU DO

On your own: read the worked example, then try these.

- E1 If $n = -11$, find $-n$ and $-(-n)$.

- E2 Is $-n$ always a negative number? Give an example.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Integers** are whole numbers: $\dots, -2, -1, 0, 1, 2, \dots$
 2. Negative numbers are below zero.
 3. The **opposite** is the same distance from 0 on the other side.
 4. 0 is its own opposite.

 Level 1

Write -7 as a real situation.

1. Negative means below or less

Answer: 7°C below zero

 Level 2

Opposite of -23 ?

1. Same distance, other side of 0

Answer: 23

 Level 3

Which integers are 4 from 0?

1. One on each side of 0

Answer: 4 and -4

 Extension

If $n = -6$, find $-n$.

1. Opposite of -6

Answer: 6

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.02

The number line

Date _____

Learning intentions

We are learning to:

- show integers on a number line
- find integers between two numbers

Success criteria

I can:

- ☐ draw a number line with an even scale
- ☐ mark positive and negative integers in the right places
- ☐ list the integers **between** two numbers

Key terms Write the definition as your teacher explains it.

Number line

Positive

Negative

Notes

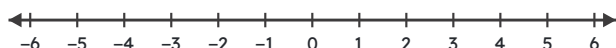
Level 1 · Getting started

Easy · 28 questions

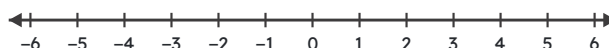
WE DO With your teacher: copy what they write on the board.

Example 1 Mark the integers on the number line.

a Mark 3, -2, 0 and -5 on the number line.



b Mark the integers between -3 and 2.



YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

28 QUESTIONS

REMEMBER Numbers **INCREASE** to the right.

ROUND 1 Questions 1-16: which integer is further to the right on the number line?

1 10 or 14

2 6 or 12

3 4 or 0

4 -11 or -3

5 0 or 3

6 3 or 12

7 3 or -1

8 -5 or 2

9 12 or -6

10 -10 or 12

11 15 or -9

12 12 or 5

13 7 or -7

14 10 or 8

15 1 or 10

16 -10 or -2

YOU DO On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 17–28: start at the first number and move along the number line. Where do you land?

17 start at 2, move 3 left

18 start at -1 , move 6 right

19 start at 7, move 4 right

20 start at 4, move 9 right

21 start at 7, move 6 right

22 start at -2 , move 5 right

23 start at -3 , move 9 right

24 start at 5, move 7 right

25 start at 1, move 7 left

26 start at -2 , move 3 left

27 start at 3, move 9 left

28 start at -4 , move 9 right

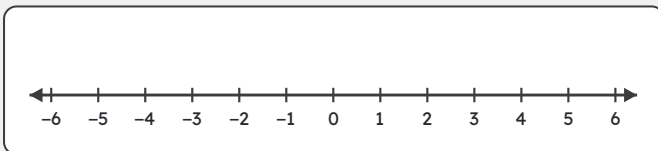
Level 2 · Building up

Medium · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Mark the integers, then answer the question.

a Mark -4 , 2 , -1 and 5 .



b Which integer is 3 to the left of -2 ?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Negatives are left of 0.

ROUND 1 Questions 1–16: how many units apart are the two integers on the number line?

1 -14 and 8

2 -1 and 1

3 -16 and 20

4 3 and 20

5 -6 and 14

6 1 and 5

7 4 and 1

8 -18 and -3

9 0 and 7

10 -10 and -2

11 -6 and 20

12 -3 and 15

13 -15 and 16

14 1 and 2

15 -15 and -1

16 5 and 18

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: which integer is exactly halfway between?

17 5 and 13

18 -10 and 4

19 -1 and 7

20 -6 and -2

21 3 and 13

22 -14 and 4

23 -11 and -5

24 -5 and 5

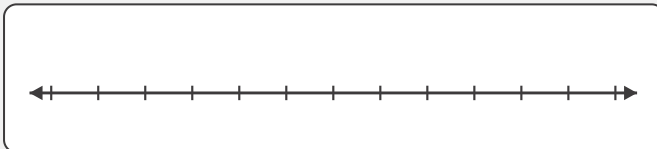
Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Choose a scale, then mark the integers.

a Mark -30, 10 and -15 using steps of 5.



b How many integers are between -6 and 4?

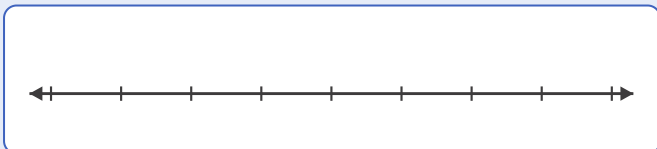
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Number lines with a scale.

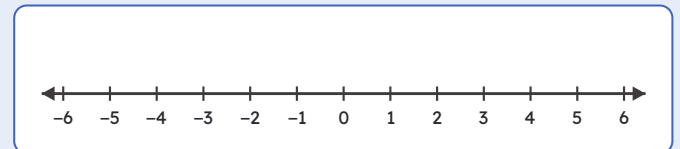
2 QUESTIONS

REMEMBER Use an **EVEN SCALE**: equal gaps.

1 Mark -20, 15, -5 and 30. Use steps of 10.



2 Mark the integers more than -4 but at most 2.



✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

List the integers between -3 and 1.

Level 2

Which is further left on a number line: -8 or -2?

Level 3

Which integer is halfway between -9 and 3?

 Extension · Stretch yourself Worked example

How many integers are between -15 and 12 ?

1. From -14 to -1 : 14 integers
2. Then 0: 1 integer
3. From 1 to 11: 11 integers

Answer: $14 + 1 + 11 = 26$ integers

YOU DO

On your own: read the worked example, then try these.

- E1 How many integers are between -30 and 25 ?

- E2 Which integer is halfway between -17 and 5 ?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Numbers **increase** to the right.
 3. Use an **even scale**: equal gaps.

2. Negatives are left of 0.
4. "Between" does not include the end numbers.

 Level 1

Where is -3 ?

1. 3 steps left of 0

Answer: 3 marks left of zero

 Level 2

Integers between -2 and 2 ?

1. Do not include -2 and 2

Answer: $-1, 0, 1$

 Level 3

Mark -20 on a scale of 10s

1. Each gap is 10
2. 2 gaps left of 0

Answer: second mark left of 0

 Extension

Halfway between -6 and 2 ?

1. Distance 8, half is 4
2. $-6 + 4$

Answer: -2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.03

Comparing and ordering integers

Date _____

Learning intentions

We are learning to:

- compare integers using $<$ and $>$
- put integers in ascending and descending order

Success criteria

I can:

- ☐ use a number line to decide which integer is larger
- ☐ write true statements with $<$ and $>$
- ☐ order a set of integers from smallest to largest and largest to smallest

Key terms Write the definition as your teacher explains it.

Ascending

Descending

$<$ and $>$

Notes

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Decide true or false, then order the integers.

a True or false: $-4 > 2$? $-2 < -3$?

b Write in ascending order: 3, -6, 0, -1, 5

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER Further **RIGHT** on the number line = **LARGER**.

ROUND 1 Questions 1-16: write $<$ or $>$ in the box.

1 $-4 \square -18$

2 $-9 \square -3$

3 $-2 \square -9$

4 $1 \square 2$

5 $8 \square -17$

6 $-14 \square 1$

7 $-20 \square -1$

8 $11 \square 1$

9 $-20 \square -4$

10 $1 \square 7$

11 $-5 \square 6$

12 $-2 \square 12$

13 $11 \square -2$

14 $-15 \square -11$

15 $9 \square -4$

16 $-6 \square 10$

YOU DO On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

ROUND 2 Questions 17–20: write the integers in descending order (largest first).

17 21, 22, 15, -10, 3

18 -26, -25, 27, -24, 12

19 19, 28, 20, 2, 13

20 24, -24, -39, -4, 28

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order the integers.

a Ascending: $-(-3)$, -5, 1, $-(2)$

b Write 3 integers between -12 and -8 in ascending order.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Challenge.

3 QUESTIONS

REMEMBER **ASCENDING**: smallest to largest.

1 Write 3 integers that are less than -5 but greater than -10.

2 Which is the smallest: $-(-6)$, -8 or $-(-10)$?

3 Put in ascending order: -1, -10, -100, -1000, 0

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

True or false: $-7 > -3$?

Level 2

Write in descending order: 2, -5, 0, -1

Level 3

Write in ascending order: -12, 5, $-(-8)$, -20, 0

 Extension · Stretch yourself Worked example

Write in ascending order: -3 , 2 , $-(-5)$, -7 , 0

- $-(-5)$ is the opposite of -5 , which is 5
- Now order: -7 , -3 , 0 , 2 , 5

Answer: -7 , -3 , 0 , 2 , $-(-5)$

YOU DO

On your own: read the worked example, then try these.

- E1 Order from smallest: $-(-2)$, -4 , $-(-4)$, 3 , $-(-9)$

- E2 Find two integers between $-(-3)$ and $-(4)$.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Further **right** on the number line = **larger**.
 - $<$ means “is less than”; $>$ means “is greater than”.
 - Ascending**: smallest to largest.
 - Descending**: largest to smallest.

 Level 1

Is $-5 < -1$?

- -5 is further left

Answer: True

 Level 2

Order -2 , 4 , -8 , 1

- Smallest is the most negative

Answer: -8 , -2 , 1 , 4

 Level 3

Descending: -3 , -30 , 3 , 0

- Largest first

Answer: 3 , 0 , -3 , -30

 Extension

Order $-(-4)$, -6 , 2

- $-(-4) = 4$

Answer: -6 , 2 , 4

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.04

Adding integers

Date _____

Learning intentions

We are learning to:

- add positive and negative integers

Success criteria

I can:

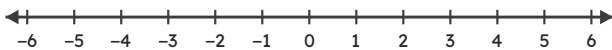
- ☐ use a number line to add integers
- ☐ know that adding a negative is the same as subtracting
- ☐ add three or more integers

Key terms Write the definition as your teacher explains it.**Sum****Adding a negative**

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate. Use the number line for part a.**a** $3 + (-4)$. Use the number line.**b** $-8 + (-2)$ and $-4 + 8$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS****REMEMBER** Adding a **POSITIVE**: move right.**ROUND 1** Questions 1–16: evaluate each sum.

1 $-3 + 8$

2 $-1 + 7$

3 $-12 + 10$

4 $-4 + 9$

5 $-10 + 12$

6 $-10 + 6$

7 $-5 + 11$

8 $-5 + 10$

9 $-6 + 6$

10 $-9 + 5$

11 $-12 + 11$

12 $-8 + 11$

13 $-7 + 10$

14 $-3 + 10$

15 $-9 + 4$

16 $-3 + 12$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 13–20: evaluate each sum.

13 $-82 + (-63)$

14 $54 + 78$

15 $38 + 87$

16 $-57 + 75$

17 $35 + 50$

18 $62 + (-56)$

19 $-54 + 79$

20 $-83 + 52$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** It is -7°C and the temperature rises 12°C . New temperature?

Your teacher works this out. Copy every step.

- b** Find the missing number: $\square + 9 = -4$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Adding integers in context.

3 QUESTIONS

REMEMBER Same signs: add and keep the sign.

- 1** The temperature is -6°C and rises 11°C . What is the new temperature?

- 2** A lift at level -2 goes up 7 floors. Which level is it at?

- 3** Maya has \$25 and spends \$40 on credit. What is her balance?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-6 + 9$

Level 2

$-4 + (-8) + 5$

Level 3

It is -3°C and the temperature drops 5°C , then rises 9°C . What is the temperature now?

 Extension · Stretch yourself Worked exampleFind the missing number: $\square + (-14) = -5$

1. Think: what plus -14 gives -5 ?
2. Undo by adding 14: $-5 + 14 = 9$
3. Check: $9 + (-14) = -5$ ✓

Answer: 9

YOU DO

On your own: read the worked example, then try these.

E1 $\square + (-8) = -20$

E2 $-13 + \square + 6 = 0$

Lesson summary

Missed the lesson? Start here.

Remember

1. Adding a **positive**: move right.
2. Adding a **negative**: move left.
3. Same signs: add and keep the sign.
4. Different signs: subtract, keep the sign of the bigger number.

 Level 1

$4 + (-6)$

1. Start at 4, move 6 left

Answer: -2

 Level 2

$-5 + (-3)$

1. Same signs: $5 + 3 = 8$, keep -

Answer: -8

 Level 3

$-9 + 4 + (-2)$

1. $-9 + 4 = -5$
2. $-5 + (-2) = -7$

Answer: -7

 Extension

$\square + (-4) = 1$

1. $1 + 4 = 5$

Answer: 5

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.05

Subtracting integers

Date _____

Learning intentions

We are learning to:

- subtract positive and negative integers

Success criteria

I can:

- ☐ subtract a positive integer by moving left
- ☐ know that subtracting a negative is the same as adding
- ☐ simplify expressions with + and – signs

Key terms Write the definition as your teacher explains it.

Difference

Subtracting a negative

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate.

a $7 - 11$ and $-4 - 8$

b $3 - (-4)$ and $-5 - (-9)$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Subtracting a **POSITIVE**: move left.

ROUND 1 Questions 1–16: evaluate. Subtract by moving left on the number line.

1 $-6 - 8$

2 $6 - 11$

3 $-7 - 4$

4 $-6 - 5$

5 $-5 - 5$

6 $-5 - 13$

7 $-5 - 14$

8 $-1 - 13$

9 $4 - 8$

10 $5 - 8$

11 $2 - 5$

12 $-2 - 8$

13 $8 - 6$

14 $-9 - 8$

15 $2 - 10$

16 $-6 - 9$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: evaluate. Work from left to right.

17 $6 - (-8) + (-7)$

18 $8 - (-11) + 3$

19 $-10 - (-9) + 13$

20 $13 - 13 + 8$

21 $-10 - (-2) + (-5)$

22 $7 - (-12) + (-15)$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** The temperature rose from -9°C to 6°C . By how much?

Your teacher works this out. Copy every step.

- b** Find the missing number: $-2 - \square = 6$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Subtraction in context.

3 QUESTIONS

REMEMBER $-(-)$ becomes $+$.

- 1 The minimum temperature was -5°C and the maximum was 11°C . By how much did it rise?

- 2 Find the difference between -18 m and 7 m .

- 3 A submarine at -120 m rises 45 m . How deep is it now?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-3 - 6$

Level 2

$4 - (-9)$

Level 3

The temperature went from -7°C to 12°C . How much did it rise?

 Extension · Stretch yourself Worked exampleFind the missing number: $6 - \square = 10$

1. Subtracting a negative adds
2. $6 + 4 = 10$, so we subtracted -4
3. Check: $6 - (-4) = 10$ ✓

Answer: -4

YOU DO

On your own: read the worked example, then try these.

E1 $-3 - \square = 5$

E2 $\square - (-8) = -2$

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Subtracting a **positive**: move left.
 2. Subtracting a **negative** = adding the positive.
 3. $- (-)$ becomes $+$.
 4. Difference between two temperatures: higher – lower.

 Level 1

$5 - 9$

1. Start at 5, move 9 left

Answer: -4 Level 2

$-2 - (-6)$

1. $= -2 + 6$

Answer: 4 Level 3

$-4 - 3 - (-8)$

1. $-4 - 3 = -7$

2. $-7 + 8$

Answer: 1 ExtensionRise from -8°C to 5°C

1. $5 - (-8) = 13$

Answer: 13°C

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LESSON
1.06

Multiplying integers

Date _____

Learning intentions

We are learning to:

- multiply positive and negative integers

Success criteria

I can:

- ☐ know the sign rules for multiplying
- ☐ multiply two integers
- ☐ multiply three integers by working in pairs

Key terms Write the definition as your teacher explains it.

Product

Sign rule

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate each product.

a $6 \times (-3)$ and -2×5

Your teacher works this out. Copy every step.

b $-8 \times (-6)$ and $-6 \times (-4) \times (-1)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER $++ = +$

ROUND 1 Questions 1–16: evaluate each product. Same signs: positive. Different signs: negative.

1 -3×6

2 $-9 \times (-1)$

3 $4 \times (-10)$

4 $6 \times (-7)$

5 $-6 \times (-5)$

6 $-8 \times (-10)$

7 -7×2

8 -6×3

9 $-8 \times (-6)$

10 $3 \times (-8)$

11 $-9 \times (-8)$

12 -5×1

13 $-3 \times (-1)$

14 -3×10

15 $2 \times (-3)$

16 -7×1

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: write the sign of the answer only: + or –.

17 $(-4) \times 3 \times (-6) \times 7$

18 5×4

19 $(-5) \times (-2)$

20 $(-7) \times (-9)$

21 $(-8) \times (-1) \times (-8)$

22 $8 \times 5 \times (-5)$

23 $3 \times 4 \times (-2)$

24 $3 \times (-8) \times 2 \times (-9)$

25 $(-6) \times 7 \times 4 \times 3$

26 $(-2) \times (-7) \times 3 \times (-5)$

27 $(-6) \times 4 \times 2 \times (-5)$

28 2×2

29 $(-3) \times (-8) \times (-5)$

30 $(-5) \times (-7)$

31 $(-7) \times 4 \times 3$

32 $5 \times (-1) \times 9 \times 5$

Level 2 · Building up

Medium · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Evaluate.

a $-2 \times 3 \times 7$

b $-4 \times 2 \times (-5)$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER $- \times - = +$

ROUND 1 Questions 1–12: evaluate. Multiply two numbers at a time.

1 $-5 \times 4 \times 5$

2 $1 \times 5 \times (-4)$

3 $-2 \times 1 \times 3$

4 $-1 \times 4 \times (-1)$

5 $-3 \times (-2) \times (-4)$

6 $-6 \times 5 \times 5$

7 $-3 \times (-6) \times (-2)$

8 $-1 \times 4 \times 5$

9 $-3 \times (-1) \times 2$

10 $2 \times (-1) \times 4$

11 $-6 \times (-1) \times (-5)$

12 $-1 \times (-2) \times (-2)$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 13–20: find the missing number.

13 $-6 \times \square = 30$

14 $2 \times \square = -2$

15 $-3 \times \square = 18$

16 $3 \times \square = 27$

17 $-3 \times \square = -15$

18 $6 \times \square = 36$

19 $-3 \times \square = -18$

20 $-5 \times \square = -10$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the missing number, or explain.

a Find the missing number: $\square \times (-7) = 56$

b Is $(-1)^{100}$ positive or negative? Explain.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER $+ \times - = -$

1 A diver descends 4 m every minute. Where is she after 7 minutes?

2 Find the missing number: $-6 \times \square = 42$

3 Is the product of 5 negative numbers positive or negative? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

-7×6

Level 2

$-8 \times (-9)$

Level 3

$-3 \times (-4) \times (-5)$

Extension · Stretch yourself**Worked example**

Evaluate $(-1) \times (-1) \times (-1) \times \dots$ with 15 factors of -1 .

1. Each pair of -1 s makes $+1$
2. 15 factors = 7 pairs and one left over

Answer: -1

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $(-2)^5$

E2 Find two integers with a product of -24 and a sum of 5 .

Lesson summary

Missed the lesson? Start here.

Remember 1. $++ = +$
3. $+- = -$

2. $-- = +$
4. Odd number of negatives $\rightarrow$ negative answer.

Level 1

$$4 \times (-7)$$

1. Different signs $\rightarrow$ negative

Answer: -28

Level 2

$$-9 \times (-3)$$

1. Same signs $\rightarrow$ positive

Answer: 27

Level 3

$$-2 \times (-5) \times (-3)$$

1. $-2 \times (-5) = 10$
2. $10 \times (-3)$

Answer: -30

Extension

$$(-3)^3$$

1. Three negatives $\rightarrow$ negative
2. 27

Answer: -27

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.07

Dividing integers

Date _____

Learning intentions

We are learning to:

- divide positive and negative integers

Success criteria

I can:

- ☐ use the same sign rules for dividing as for multiplying
- ☐ divide two integers
- ☐ work left to right with two divisions

Key terms Write the definition as your teacher explains it.**Quotient****Sign rule**

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate each quotient.

a $-12 \div (-4)$ and $15 \div (-5)$

Your teacher works this out. Copy every step.

b $24 \div (-6) \div (-2)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS****REMEMBER** Same signs $\rightarrow$ positive.**ROUND 1** Questions 1–16: evaluate each quotient. Same signs: positive. Different signs: negative.

1 $-60 \div (-10)$

2 $-30 \div 5$

3 $-10 \div 10$

4 $-12 \div (-3)$

5 $14 \div (-2)$

6 $-72 \div (-9)$

7 $45 \div (-5)$

8 $-8 \div 2$

9 $25 \div (-5)$

10 $21 \div (-3)$

11 $-36 \div (-6)$

12 $-30 \div (-10)$

13 $15 \div (-5)$

14 $-10 \div 1$

15 $-35 \div 5$

16 $-6 \div (-1)$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 13–20: find the missing number.

13 $14 \div \square = -7$

14 $45 \div \square = -9$

15 $27 \div \square = -9$

16 $32 \div \square = -4$

17 $14 \div \square = 7$

18 $-4 \div \square = 4$

19 $-72 \div \square = -8$

20 $4 \div \square = 1$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** A debt of \$96 is shared by 8 people. Write each share as an integer.

Your teacher works this out. Copy every step.

- b** Find the missing number: $\square \div (-6) = -9$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Division in context.

3 QUESTIONS

REMEMBER 0 divided by any number is 0.

- 1 A \$-84 debt is shared equally by 4 people. What is each share?

- 2 The temperature fell 18°C over 6 hours. What was the change each hour?

- 3 Find the missing number: $\square \div (-5) = 7$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-36 \div 9$

Level 2

$-42 \div (-6)$

Level 3

$-80 \div (-4) \div (-5)$

 Extension · Stretch yourself Worked exampleEvaluate $-6 \times 8 \div (-12)$

1. Work left to right
2. $-6 \times 8 = -48$
3. $-48 \div (-12) = 4$

Answer: 4

YOU DO

On your own: read the worked example, then try these.

E1 $-90 \div (-3) \times (-2)$

E2 $-5 \times (-12) \div (-4) \div 3$

Lesson summary

Missed the lesson? Start here.

Remember

1. Same signs $\rightarrow$ positive.
2. Different signs $\rightarrow$ negative.
3. 0 divided by any number is 0.
4. Two divisions: work left to right.

 Level 1

$-20 \div 4$

1. Different signs

Answer: -5

 Level 2

$-35 \div (-7)$

1. Same signs

Answer: 5

 Level 3

$-40 \div (-2) \div (-5)$

1. $-40 \div (-2) = 20$
2. $20 \div (-5)$

Answer: -4

 Extension

$\square \div (-3) = -6$

1. $-6 \times (-3)$

Answer: 18

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LESSON
1.08

Order of operations with integers

Date _____

Learning intentions

We are learning to:

- use the order of operations with positive and negative numbers

Success criteria

I can:

- ☐ do brackets first, then $\times$ and $\div$, then $+$ and $-$
- ☐ work left to right for operations of the same kind
- ☐ evaluate an expression with a fraction bar

Key terms Write the definition as your teacher explains it.**Order of operations****Fraction bar**

Notes

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate. Use the order of operations.

a $-6 + (-3) \times 2$

Your teacher works this out. Copy every step.

b $(9 - 17) \div (-4 + 6)$, written as a fraction

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS****REMEMBER** Brackets first.**ROUND 1** Questions 1–12: evaluate. $\times$ and $\div$ before $+$ and $-$.

1 $1 + (-6) \times (-1)$

2 $10 - 5 \times (-5)$

3 $12 + 6 \times (-4)$

4 $12 - (-4) \times (-1)$

5 $-2 + (-4) \times (-2)$

6 $10 + (-2) \times 2$

7 $11 - 2 \times 5$

8 $5 + (-2) \times 1$

9 $-9 + 5 \times (-4)$

10 $-11 - (-6) \times (-5)$

11 $-5 + 4 \times 6$

12 $-8 + 3 \times (-3)$

YOU DO On your own: **Medium** keep going: Round 2.

3 QUESTIONS MORE

ROUND 2 Questions 13–15: evaluate. Show each step.

13 $-7 - (-5 + (-2)) \times (-3)$

14 $8 - (-5 + (-2)) \times 2$

15 $-6 - (7 + (-1)) \times (-4)$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Show every step.

a $\frac{7 + (-12) + 6}{2 - 5}$

Your teacher works this out. Copy every step.

b $-6 + (-6) \div (-2) \times 2$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Fraction bars: work out the top and bottom first.

3 QUESTIONS

REMEMBER + and – from left to right.

1 $\frac{9 - 17}{-4 + 6}$

2 $\frac{7 + (-12) + 6}{2 - 5}$

3 $\frac{-3 \times 8}{-6}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-5 + 2 \times 3$

Level 2

$-3 \times (4 - 9)$

Level 3

$\frac{-4 \times 6}{3 - 5}$

Extension · Stretch yourself**Worked example**

Insert brackets to make this true: $-2 + 6 \times -3 = -12$

1. Try $(-2 + 6) \times -3$
2. $= 4 \times (-3) = -12$ ✓

Answer: $(-2 + 6) \times -3$

YOU DO

On your own: read the worked example, then try these.

- E1** Insert brackets: $8 - 12 \div -4 + 2 = 2$

- E2** Write an expression using -3 , 4 and 5 that equals -17 .

Lesson summary

Missed the lesson? Start here.

Order of operations

1. Brackets first.

3. $+$ and $-$ from left to right.

2. $\times$ and $\div$ from left to right.

4. Fraction bar: work out top and bottom first.

Level 1

$$-5 + 4 \times (-2)$$

1. $4 \times (-2) = -8$
2. $-5 + (-8)$

Answer: -13

Level 2

$$(-3 - 5) \times 2$$

1. $-3 - 5 = -8$
2. -8×2

Answer: -16

Level 3

$$\frac{-18}{2 - 5}$$

1. $2 - 5 = -3$
2. $-18 \div (-3)$

Answer: 6

Extension

$$(-2 + 6) \times -3$$

1. 4×-3

Answer: -12

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.09

Integers on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with integers

Success criteria

I can:

- ☐ enter a negative number using the $(-)$ key
- ☐ use brackets on a calculator
- ☐ check that an answer is reasonable

Key terms Write the definition as your teacher explains it.

$(-)$ key

Estimate

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use your calculator to evaluate.

a Use your calculator: $28 - 37$ and $-17 + 11$

b $[5 + (-20)] \div (-3)$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Use the $(-)$ key for negative numbers.

ROUND 1 Questions 1–16: use your calculator. Use the $(-)$ key for negatives.

1 $-199 + 166$

2 $215 + 182$

3 $231 + 238$

4 $-212 + 45$

5 $394 + 273$

6 $184 + (-165)$

7 $-297 + 352$

8 $216 + (-296)$

9 $166 + 47$

10 $-119 + 216$

11 $-226 + 54$

12 $16 + 341$

13 $-109 + (-265)$

14 $316 + 260$

15 $-31 + 335$

16 $400 + (-381)$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: use your calculator.

17 12×38

18 -10×17

19 -3×11

20 $14 \times (-12)$

21 $14 \times (-17)$

22 7×22

23 -40×10

24 $-41 \times (-30)$

25 17×8

26 $41 \times (-6)$

27 $45 \times (-34)$

28 $-37 \times (-20)$

29 $-14 \times (-23)$

30 -42×5

31 24×10

32 $-21 \times (-44)$

Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use your calculator. Enter the brackets.

a $18 - 15 \times 2$ on a calculator

Your teacher works this out. Copy every step.

b $100 - 120 \div 4 + 17$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Calculators follow the order of operations.

ROUND 1

Questions 1–16: use your calculator.

1 $1\,782 \div 33$

2 $-24 \div 24$

3 $1\,798 \div 31$

4 $-204 \div (-17)$

5 $-1\,740 \div 30$

6 $-945 \div 35$

7 $-504 \div (-12)$

8 $62 \div 31$

9 $-624 \div (-26)$

10 $448 \div (-28)$

11 $-187 \div 11$

12 $92 \div 23$

13 $75 \div (-15)$

14 $-368 \div 23$

15 $798 \div 38$

16 $40 \div (-8)$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: use your calculator. Enter the brackets.

17 $(-27 - (-22)) \times (-17)$

18 $(2 - (-25)) \times 15$

19 $(-30 - 63) \times (-24)$

20 $(-5 - (-38)) \times 9$

21 $(-19 - (-78)) \times 17$

22 $(62 - (-16)) \times (-13)$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use your calculator.

a $-16 \times 3 + (-4)$

b $\frac{-84 - 16}{-25}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Estimate first, then check with a calculator.

3 QUESTIONS

REMEMBER Use brackets for fraction bars.

1 Estimate -49×21 , then calculate.

2 Estimate $-612 \div 29.8$ (1 d.p.), then calculate.

3 Jo typed -6^2 and got -36 . Why? How do you get 36?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-84 + 39$

Level 2

$-15 \times (-7) - 100$

Level 3

$(-35 - 45) \div (-16)$

Extension · Stretch yourself**Worked example**

Evaluate $\frac{-48 + 12}{-3 \times (-4)}$ on a calculator.

1. Put brackets around the top and the bottom
2. $(-48 + 12) \div (-3 \times (-4))$
3. $= -36 \div 12$

Answer: -3

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{-125 - 25}{6 \times (-5)}$

E2 $\frac{(-9)^2 - 1}{-16}$

Lesson summary

Missed the lesson? Start here.

Remember 1. Use the $(-)$ key for negative numbers.
3. Use brackets for fraction bars.

2. Calculators follow the order of operations.
4. Estimate to check your answer.

Level 1

$$-23 + 9$$

1. $(-) 23 + 9 =$

Answer: -14

Level 2

$$18 - 15 \times 2$$

1. $\times$ first

Answer: -12

Level 3

$$(-36 \div 6) \div 5$$

1. Brackets on top

Answer: -6

Extension

$$(-3)^2 \text{ vs } -3^2$$

1. $(-3)^2 = 9$, $-3^2 = -9$

Answer: 9 and -9

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.10

Integer problems

Date _____

Learning intentions

We are learning to:

- solve real problems using integers

Success criteria

I can:

- ☐ decide which operation a problem needs
- ☐ write an integer calculation for a word problem
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Rise

Balance

Deposit

Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Write a number sentence, then answer.

- a** The minimum overnight temperature in Goulburn was -5°C . During the day it rose to 11°C . By how much did it rise?

Your teacher works this out. Copy every step.

- b** A bank account has \$120. Three withdrawals of \$55 are made. What is the balance?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **16 QUESTIONS****REMEMBER** Read carefully and underline the numbers.**ROUND 1** Questions 1–8: find the new temperature.

- 1** It is -11°C . The temperature rises 12°C .

- 2** It is -3°C . The temperature rises 3°C .

- 3** It is 13°C . The temperature rises 13°C .

- 4** It is 7°C . The temperature falls 13°C .

- 5** It is -7°C . The temperature rises 11°C .

- 6** It is 4°C . The temperature rises 12°C .

- 7** It is 7°C . The temperature rises 5°C .

- 8** It is 2°C . The temperature rises 15°C .

YOU DO On your own: **Medium** keep going: Round 2.

2 QUESTIONS MORE

ROUND 2 Questions 9–10: write a number sentence, then answer.

- 9 A diver is at -80 m. She rises 3 m a minute for 4 minutes. Where is she now?

- 10 A diver is at -95 m. She rises 2 m a minute for 2 minutes. Where is she now?

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write a number sentence, then answer.

- a Harrison saves \$80 a week and pays \$210 a month for a car loan. How much after 6 months (26 weeks)?

Your teacher works this out. Copy every step.

- b A diver at -42 m rises 6 m every minute. Where is she after 5 minutes?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Multi-step problems.

2 QUESTIONS

REMEMBER Difference: larger – smaller.

- 1 Harrison saves \$80 a week. Each month \$210 is taken out for a car loan. How much is in the account after 6 months (26 weeks)?

- 2 A game scores $+5$ for a win and -3 for a loss. Mia wins 7 and loses 9. What is her score?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

It was -6°C and rose 10°C . What is the temperature now?

Level 2

A diver goes from -30 m to -12 m. How far did she rise?

Level 3

A score starts at 0. Gain 4, lose 7, lose 7, gain 3. Final score?

Extension · Stretch yourself**Worked example**

A submarine at -240 m rises 15 m per minute. How long until it is at -60 m?

1. Distance to rise: $-60 - (-240) = 180$ m
2. Time: $180 \div 15 = 12$

Answer: 12 minutes

YOU DO

On your own: read the worked example, then try these.

- E1** A freezer is at 4°C and cools by 3°C every 10 minutes. How long until it is -17°C ?

- E2** Write your own integer problem with an answer of -8 .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read carefully and underline the numbers.
 2. Rise, gain, deposit $\rightarrow$ add. Fall, loss, withdraw $\rightarrow$ subtract.
 3. Difference: larger $-$ smaller.
 4. Check the answer makes sense.

Level 1

From -5°C to 11°C

1. $11 - (-5)$

Answer: 16°C rise

Level 2

$\$20$ balance, withdraw $\$50$

1. $20 - 50$

Answer: $-\$30$

Level 3

6 months: $+\$80 \times 26$, $-\$210 \times 6$

1. $2080 - 1260$

Answer: $\$820$

Extension

Rise 15 m/min from -240 m to -60 m

1. $180 \div 15$

Answer: 12 minutes

This space is left blank on purpose. It is the back of your exit ticket.